

AURA FRAUD INVESTIGATION REPORT

AI-Assisted Transaction Investigation

1. Transaction Overview

Transaction ID	b991672c504a79471a46d05f02ca4109
Risk Band	LOW
Risk Score	14.86
Fraud Probability	100.00%
Transaction Amount	9.11
Investigation Flag	False
Recommended Action	APPROVE

2. Executive Summary

Transaction b991672c504a79471a46d05f02ca4109 has been classified as LOW risk with an AURA risk score of 14.86. No immediate investigation is indicated by the current evidence.

3. Risk Interpretation

- The AURA risk engine has not flagged this transaction for investigation.
- The machine-learning fraud probability is 100.00%.
- The transaction amount is 9.11.

4. Investigation Evidence

Evidence Type	Description
investigation_reason	Model identified an anomalous transaction pattern

5. Top Risk Factors

Feature	Observed Value	SHAP Value
amount_deviation_ratio	0.1345828368826214	0.051617092143257735
log_amount	2.3135250330323798	0.04619191242869194
amt	9.11	0.04276156021109219
hour	10	0.040466517800828015
category_seen_before	130	0.03903113855915486

6. Graph Analysis

Metric	Value
Graph Risk Score	14.295539263211863
Graph Risk Band	LOW
Transaction Count	1535
Fraud Transaction Count	11
Fraud Ratio	0.0071661237785016286

Graph Reasons:

- Customer is connected to previously identified fraudulent transaction(s)

7. AI Investigation Analysis

AI Assessment

The transaction (ID b991672c504a79471a46d05f02ca4109) has been classified as LOW risk with a risk score of 14.86 and an absolute fraud probability of 1.0. The deterministic report indicates that no immediate investigation is warranted, and the recommended action is APPROVE.

Key Fraud Indicators (Evidence)

Indicator	Evidence	Interpretation
Anomaly detection	`anomaly_score`: -0.072 - flagged as "Model identified an anomalous transaction pattern"	Anomaly score is an investigative signal, not a fraud verdict.
Behavioural deviation	`amount_deviation_ratio`: 0.1346; `log_amount`: 2.31; `hour`: 10	These features contributed positively to the risk score (SHAP values 0.04-0.05).
Graph risk	`graph_risk_score`: 14.30; `graph_reasons`: "Customer is connected to previously identified fraudulent transaction(s)"	The connection to known fraud raises the graph risk but remains in the LOW band.
Transaction context	`amount`: \$9.11; `customer_prev_avg_amount`: \$67.69; `customer_prev_std_amount`: \$150.08	Small absolute amount, but slightly below customer mean.

Evidence Interpretation

1. Risk Score vs. Fraud Probability - The model outputs a 100 % fraud probability, yet the overall risk score remains low because behavioural and graph signals are muted by the small transaction amount and the customer's long transaction history (1,288 prior transactions).

2. Anomaly vs. Fraud - Anomaly detection highlights unusual patterns; however, the anomaly score here is modest and the transaction's features (time of day, category familiarity, merchant familiarity) align with historical behaviour.

3. Graph Connection - The customer is linked to a prior fraudulent transaction, which elevates the graph risk score. Nevertheless, the overall graph risk band is LOW, suggesting that the link does not yet cross the threshold for mandatory investigation.

Investigation Considerations

- Validate Fraud Probability - A 1.0 probability warrants review; confirm that the model is not over-fitted to this case.
- Graph Context - Investigate the nature of the connected fraudulent transaction(s): same merchant, same card, or related account?
- Historical Consistency - Compare this transaction's characteristics with the 1,288 historical transactions to ensure no subtle drift.
- Review Anomaly Contributors - Examine which feature(s) most influenced the anomaly score; consider if the `amount\_deviation\_ratio` of ~13 % could be a benign variance.
- Potential False Positive - Low amount and high fraud probability may indicate a model bias; cross-check with other low-value, low-risk transactions.

Recommended Next Steps

1. Passive Monitoring - Keep the transaction in the low-risk queue but flag it for audit if similar patterns recur.
2. Graph Audit - Run a quick review of the linked fraudulent transaction(s) to assess whether the connection is coincidental or indicative of a broader fraud ring.
3. Model Review - Schedule a model audit to confirm that the fraud probability threshold remains appropriate for low-value transactions.
4. Documentation - Log the anomaly and graph risk findings for future trend analysis.

No active investigation or hold is required at this time, consistent with the deterministic report and the supplied evidence.

8. Investigator Conclusion

The currently available evidence does not indicate a need for immediate investigation.